

Ziheng Wang

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EDUCATION

Shanghai Jiao Tong University (SJTU)

Sep. 2023 – Jun. 2028 (Expected)

B.S. in Automation

- GPA: 4.00/4.30 (Rank: 6/103)
- **Related Courses:** Robotics, Artificial Intelligence, Pattern Recognition, Digital Image Processing, Modern Control Theory, Linear Algebra, Probability and Statistics, Calculus, Signals and Systems, Embedded System and Interface Technology
- **National Scholarship** (awarded to 0.2% of undergraduates nationwide), 2024.
- **SJTU Junzheng Scholar** (50 undergraduates selected university-wide annually), 2026.

RESEARCH EXPERIENCE

EgoMatrix: Physically Faithful Dexterous Robot Demonstrations from Human Videos Apr. 2026 – Present

Advisor: Prof. Yao Mu, School of Computer Science, SJTU

- Developed an end-to-end pipeline converting egocentric human videos into dexterous robot trajectories and **physically faithful robot observations**, where depicted contacts and object motion are consistent with executed robot trajectories. Achieved SOTA on most reconstruction metrics, substantial gains in transfer success rate, and validated real-robot replay on dual-UR5e/XHand.
- Recovered the hand-object interaction from monocular video using *multi-start pose refinement*, with contact cues providing multiple starting poses for optimization. Applied *local contact refinement* to close contact gaps and reduce interpenetration, then transferred the reconstructed manipulation to dexterous robots via retargeting and reinforcement learning.
- Generated robot observations by jointly rendering the robot hands and manipulated objects from successful simulation rollouts and compositing them into the inpainted source background after removing the human hands, arms, and real-world manipulated objects; support-plane alignment and target-view rendering extend the same manipulation task to diverse backgrounds.
- Led the project as **sole first author**, responsible for method design, pipeline implementation and real-robot experiments.
- Submitted to **ICLR 2027** (*under review*). [Project page](#).

Reinforcement Learning for Multi-Robot Persistent Monitoring

Jul. 2025 – May 2026

Advisors: Prof. Xiaoming Duan and Jianping He, School of Automation and Intelligent Sensing, SJTU

- Studied multi-robot persistent monitoring, establishing a *provably equivalent event-driven MDP* and RL-based methods for minimizing worst-case weighted latency. Built **M2Bench**, the first unified benchmark platform for multi-robot persistent monitoring that supports both heuristic and RL-based methods.
- Developed three learning components: *Trans-GAE* handles asynchronous event transitions by computing each robot's advantages only across its active decision steps; Laplacian positional encodings inject graph structure into observations; and an *imitation-learning warm start* initializes RL from heuristic demonstrations via behavior cloning to reduce exploration burden.
- Designed M2Bench as a modular platform integrating 10 MDP formulations, 14 heuristic policies, and 10 RL/MARL algorithms; standardized interfaces for researchers to implement and evaluate their new MDP formulations and algorithms seamlessly.
- As **second author**, led the RL method development, M2Bench implementation, and experimental evaluation.
- Submitted to **IJRR** (*under review*). [arXiv preprint](#).

FinsROV: Low-Cost Open-Source Underwater Robot Platform

Oct. 2024 – Jul. 2025

Advisor: Prof. Jianping He, School of Automation and Intelligent Sensing, SJTU

- Built FinsROV from scratch, a low-cost open-source underwater robot with eight-thruster omnidirectional motion and onboard sensing for underwater robotics research.
- As a **core contributor**, developed the onboard control system using pressure sensing for depth and AHRS feedback for attitude, with PID control for depth, heading, and roll. Implemented the full low-level loop from sensing to thruster actuation on the physical robot.
- Placed **5th/40** in the AUV track of the 2025 China International Marine Underwater Robot Competition. [Open-source platform](#).

SKILLS

Programming: Python, C++, MatLab

Frameworks & Tools: Isaac Lab/Gym, PyTorch, OpenCV, Gymnasium, ROS/ROS2, PX4, Gazebo, Solidworks, Docker

Research interests: robot learning, dexterous manipulation, scalable robot data generation from human videos, cross-embodiment representations for manipulation, multi-robot systems